

Change Data Capture

 systemdesignschool.io/fundamentals/change-data-capture



A search feature needs to stay synchronized with the product database. Every product update—new price, changed description—should appear in search.

The naive approach: application code writes to both systems. One day, the search service times out. The database succeeds but search misses thousands of products. Data diverges.

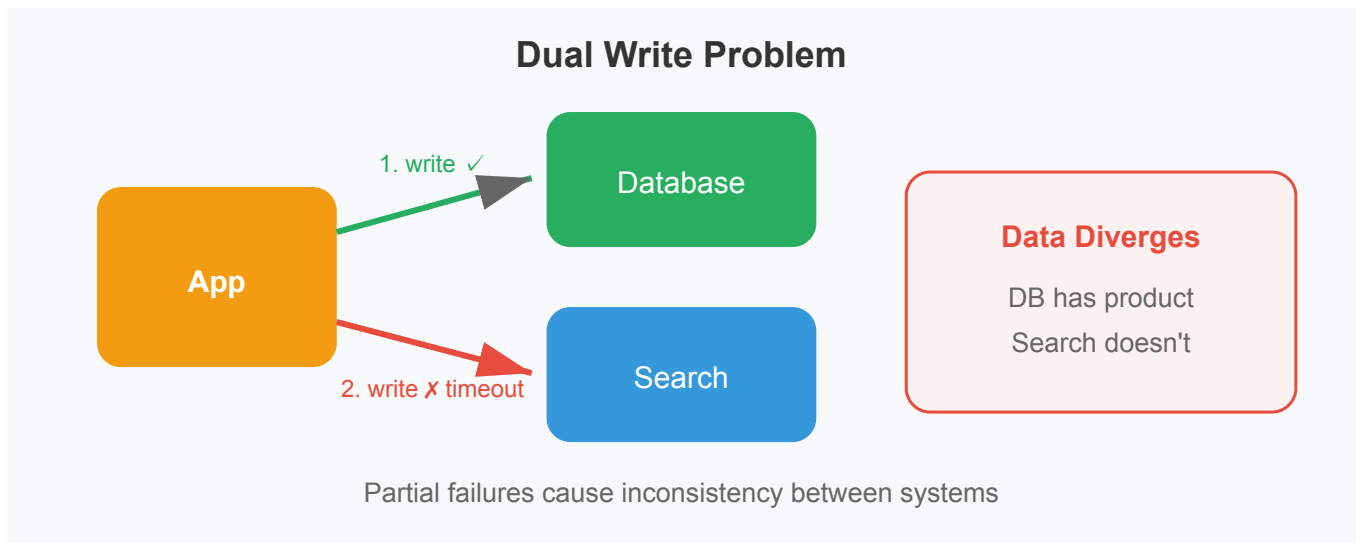
The Dual Write Problem

Application code writing to multiple systems has failure modes:

Partial failures. Database write succeeds, search write fails. Now the product exists in the database but not in search.

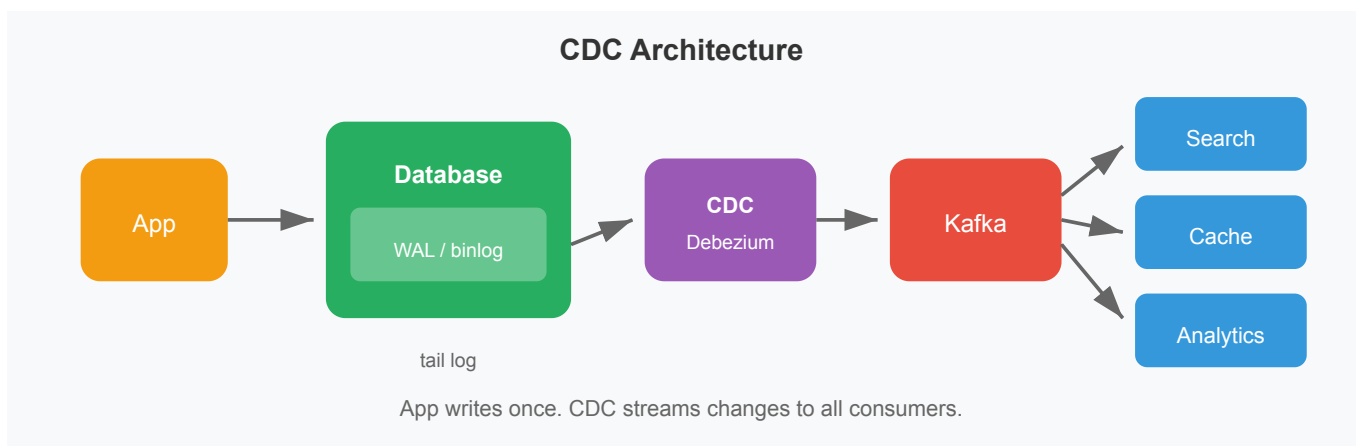
Race conditions. Two updates arrive at the database in order $A \rightarrow B$, but reach search as $B \rightarrow A$. Final states don't match.

Tight coupling. Every write path must know about every downstream system. Adding analytics means changing application code.



CDC reads the database's transaction log and streams changes to downstream systems.

The database already maintains a transaction log (MySQL binlog, Postgres WAL) for durability. CDC tools tap into this existing log.



The architecture:

1. **Database** writes changes to its transaction log
2. **CDC connector** (Debezium) tails the log, publishes events
3. **Message broker** (Kafka) stores events durably
4. **Consumers** update search, cache, analytics

Why CDC Works Better

Decoupling. Application code doesn't know about downstream systems. Adding a consumer means adding a subscription, not changing writers.

Ordering. Changes flow in the same order they occurred. No race conditions.

Completeness. Every committed change appears in the log. No events lost to partial failures.

Efficiency. Tailing a log is cheap. Only modified rows stream—no polling, no full scans.

Trade-offs

CDC adds infrastructure: connectors, brokers, consumers. For a simple app with one downstream system, dual writes with reconciliation might suffice.

CDC becomes essential when:

- Multiple downstream consumers
- Ordering matters
- You can't modify application code

Common Tools

Debezium is the most popular open-source CDC connector. It supports MySQL, Postgres, MongoDB, and others. It publishes to Kafka.

AWS DMS provides managed CDC for AWS databases.

Fivetran/Airbyte include CDC connectors for warehouse loading.

That leads into **database partitioning**—how to split data across shards when a single replica set is no longer enough.